Open in GitHub Codespaces

GitHub Codespaces

Introduction to the Codespaces environment

This guide explains how to launch and operate the desktop-enabled Codespace that powers the NGS practicals. It replaces the legacy Gitpod instructions used in previous years.

GitHub Codespaces provides a fully managed, cloud-hosted development environment that mirrors the tooling we use in class. By default, verified GitHub Global Campus students receive up to **180 core-hours per month** (for example, 90 hours on a 2-core machine) and **20 GB of storage** across active codespaces—plenty for this module's workloads. If you ever need more, you can adjust machine sizes or delete unused spaces from the Codespaces dashboard.

To deepen your understanding of Codespaces, explore these resources:

- Codespaces documentation home
- Quickstart: Use GitHub Codespaces for the first time
- Codespaces video walkthroughs
- Developing in Codespaces guide

Launching the Codespace from GitHub Web

1. Visit the repository at FairTeach/codespaces_NGS.
2. Select `<> Code` → `Create codespace on main` (or reuse an existing Codespace from the Codespaces dashboard).
3. Wait for the dev container build to complete. The process provisions Xfce, TigerVNC/noVNC, Miniforge+mamba, IGV desktop, and course-specific tooling.
4. Once VS Code connects, the desktop services start automatically under Supervisor. No additional setup is required.

> **Tip:** Reopen an existing Codespace for subsequent sessions to keep intermediate files. Starting a fresh Codespace restores the repository to its committed state.

Launch the Codespace from Manuals and PDFs

Each exported manual and PDF includes two handy badges so you can jump into Codespaces directly from the document:

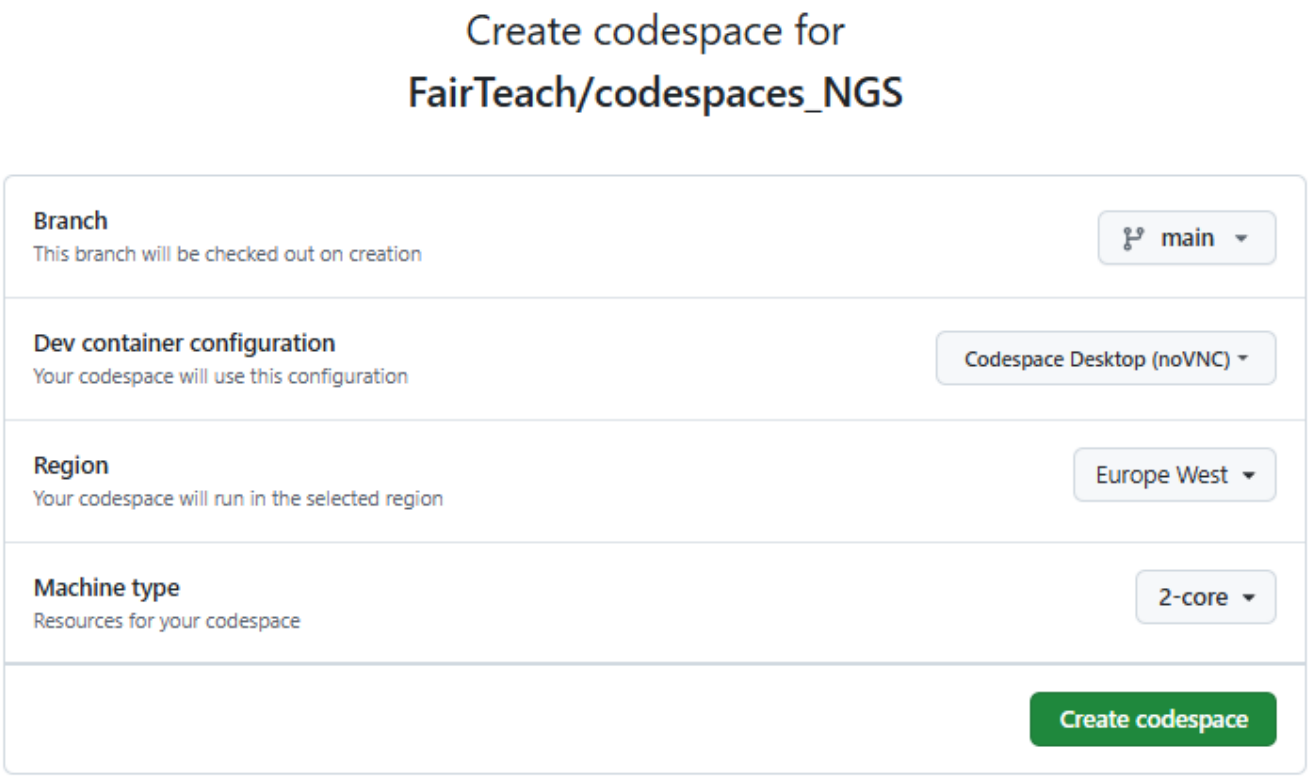
Open in GitHub Codespaces

Use **Open in GitHub Codespaces** to create a brand-new workspace from the latest commit. This gives you a clean environment if you want to restart an exercise or if your previous Codespace has expired.

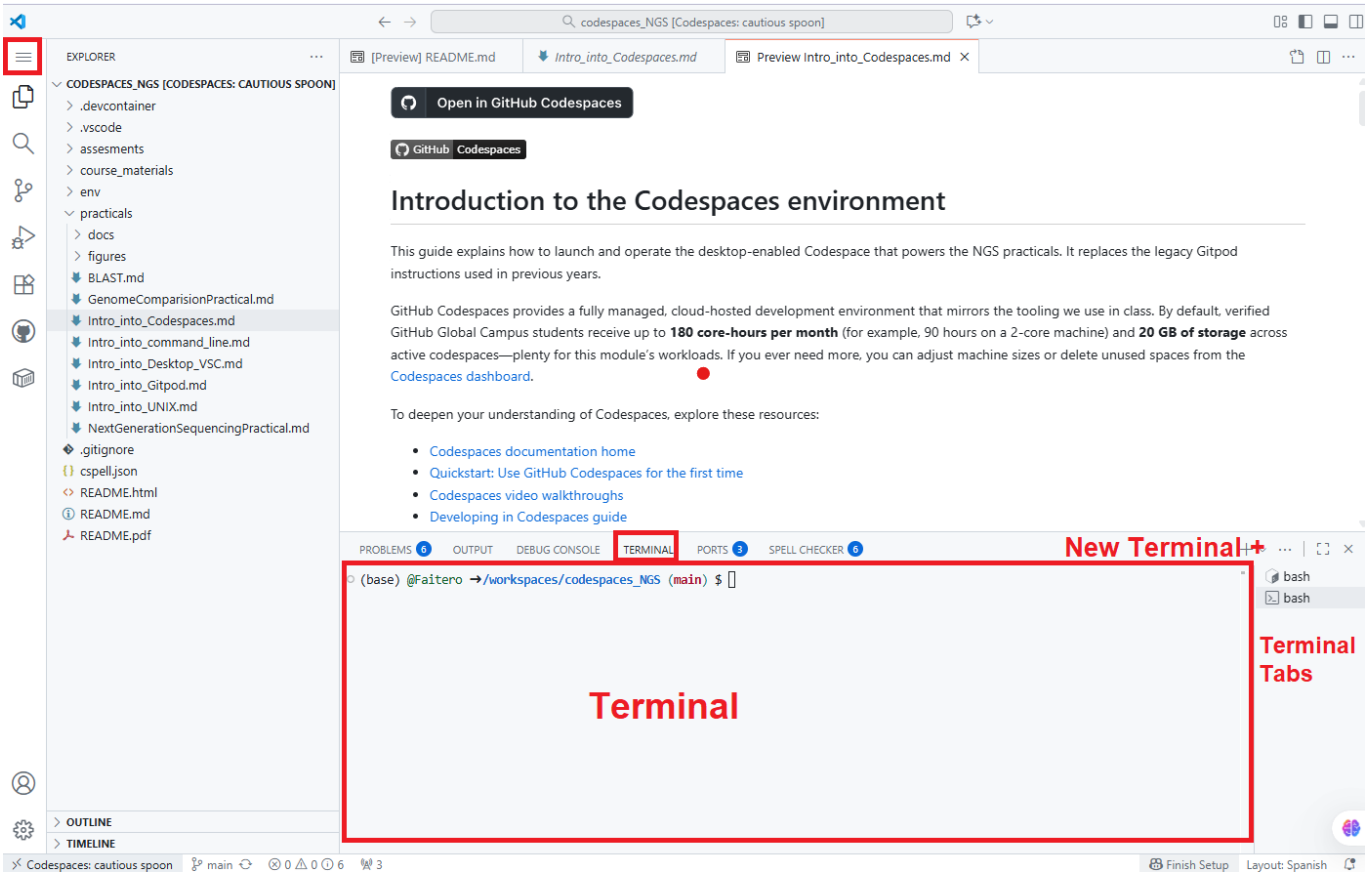
GitHub Codespaces

Choose **Continue GitHub Codespaces** to open your dashboard and resume an existing environment. Resuming keeps every file and intermediate result you produced earlier—bookmark the dashboard for quick access.

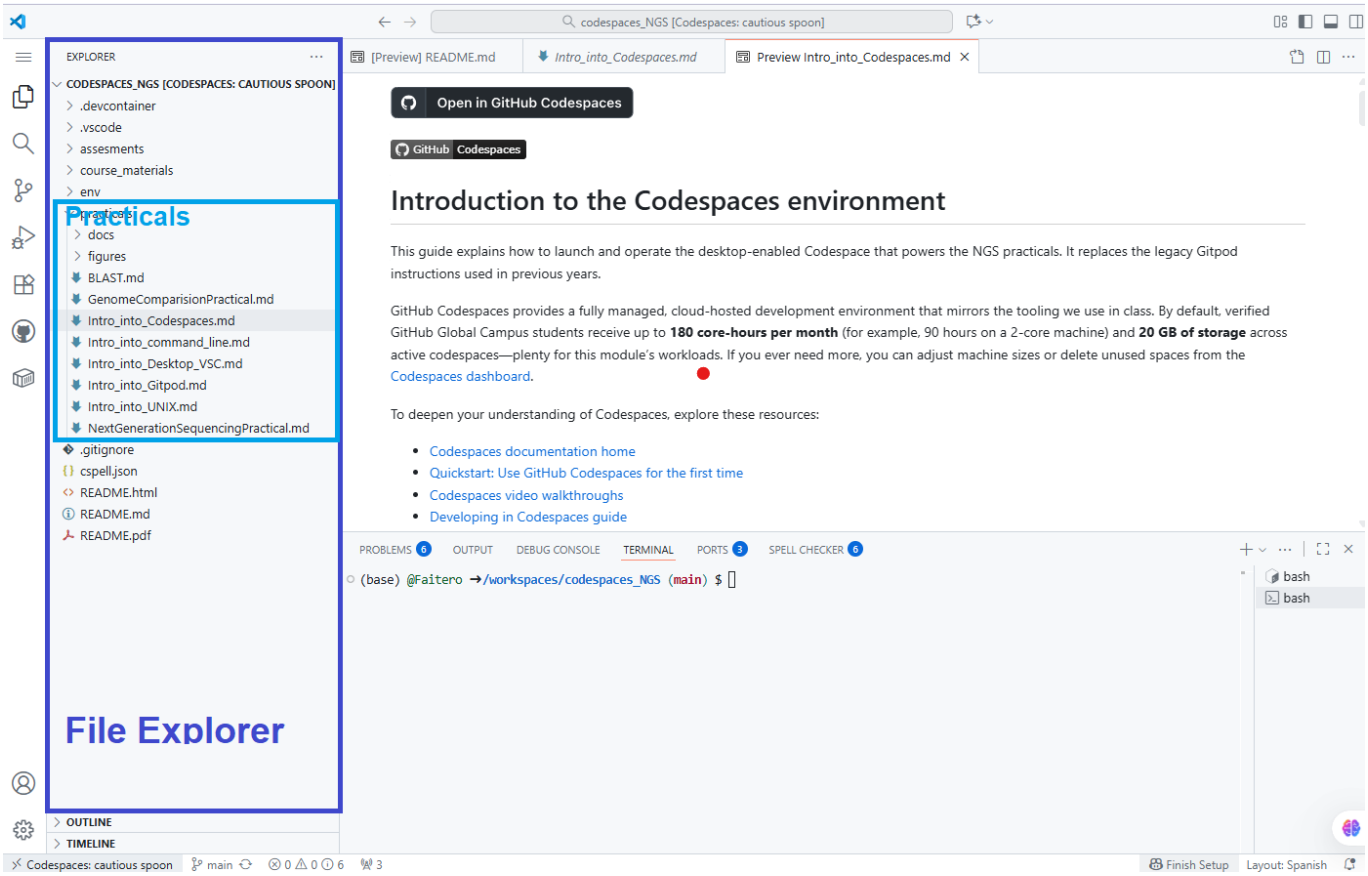
After launching a new Codespace via the badge, sign in to GitHub if prompted and confirm the default machine selection.



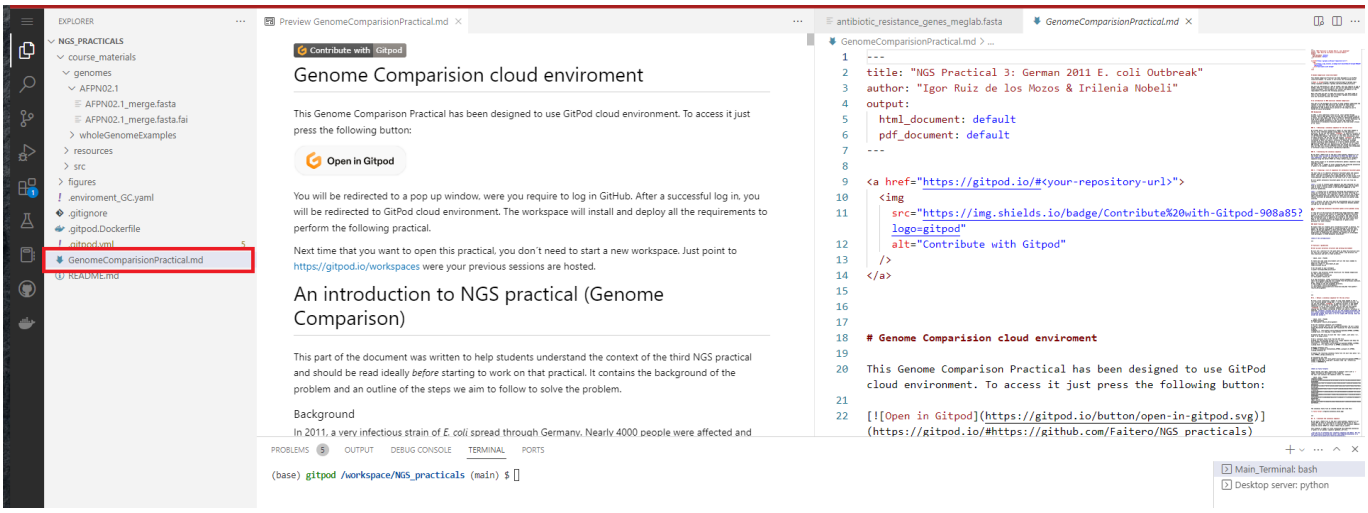
Once inside the desktop, start a terminal from `Applications → Terminal Emulator`, or continue working in the VS Code terminal.



Browse the repository through the left-hand `File Explorer`.

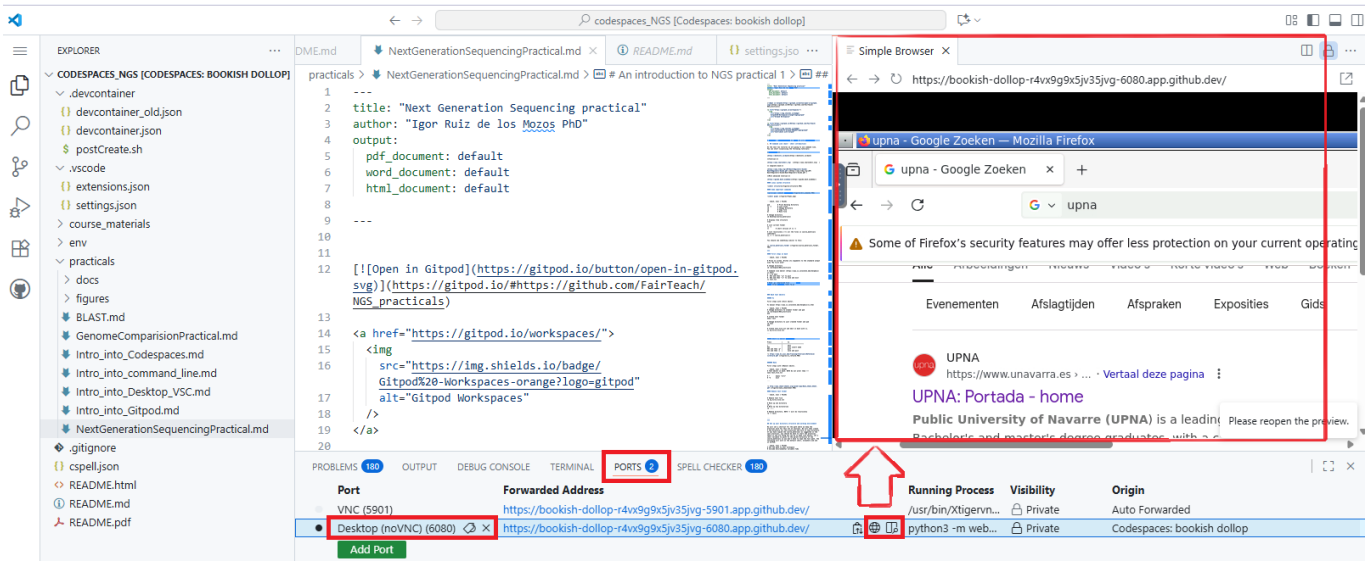


Right-click any practical Markdown (for example, `practicals/Intro_into_Codespaces.md`) and choose **Open Preview** to read the instructions alongside your terminal.



Accessing the remote desktop

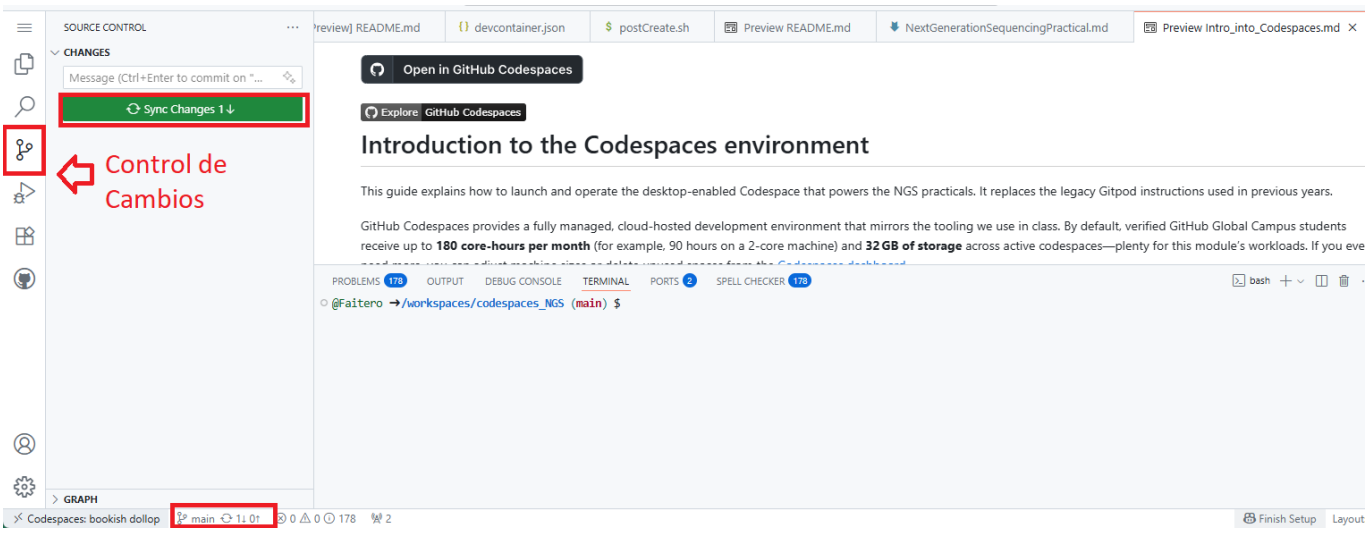
When the build finishes, Codespaces forwards the desktop ports automatically. Open the notification for `Desktop (noVNC)` on port 6080 and select **Open in Browser** to reach the Xfce desktop.



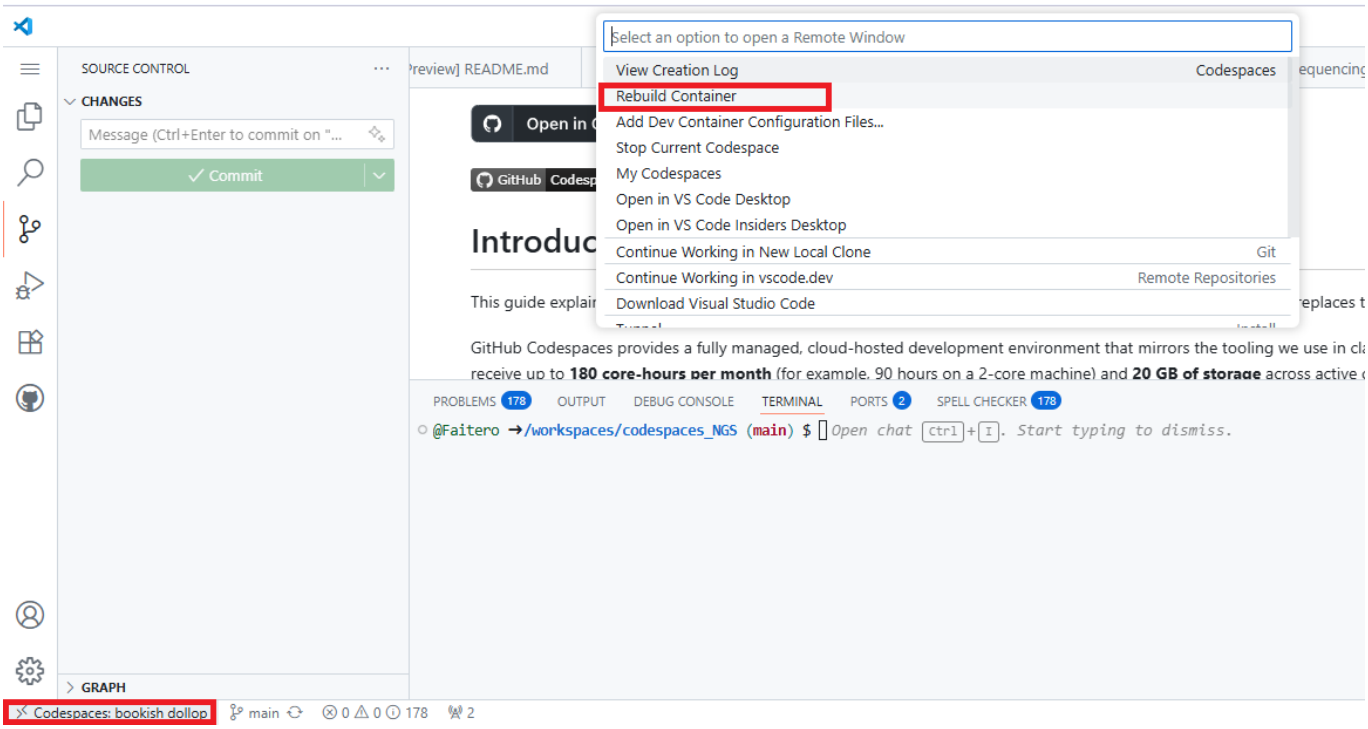
1. Open the **Ports** panel and locate port `6080` labelled `Desktop (noVNC)`.
2. Click the globe icon to open the noVNC session in a new browser tab.
3. Enter the VNC password `vscode` when prompted.

Staying in sync with the repository

- Open the **Control de cambios** / **Source Control** tab (the branch icon on the Activity Bar) to review pending edits, stage files, and make commits without leaving the editor. The view lists modified or untracked files exactly as `git status` would.
- Prefer the terminal? Run `git status` to inspect pending work, then pull updates with `git pull origin main` (or whatever branch you are tracking) to stay current with the course repository.



- After pulling, look for a notification on the bottom-left **Codespaces** tab (two stacked squares). From that menu, choose **Rebuild Container** anytime files under `.devcontainer/` change or the environment behaves unexpectedly. You can reach the same command via `F1 → Codespaces: Rebuild Container`. Wait for the rebuild to complete before resuming work.



Working with conda/mamba

- `mamba` is preinstalled; run `which mamba` or `mamba --version` to verify.
- Shells automatically source Miniforge (`/opt/miniforge/etc/profile.d/conda.sh`), so the base environment is active whenever you open a new terminal.
- Install extra packages with `mamba install -y <package>` (for example, `mamba install -y samtools`) and confirm with the corresponding `--version` command.

Running IGV

- Launch IGV desktop from any terminal with `igv &`; the application opens inside the noVNC desktop.
- Load data via `File → Load Genome from Server` or by browsing to files stored in the workspace.

For the complete repository overview and module links, see the root `README.md`.

You will also find the **Continue GitHub Codespaces** badge at the end of every manual as a quick way to reopen your existing workspace: GitHub Codespaces